ASSIGNMENT 1

1) What is data, and why is it important in the modern world?

Data means a collection of facts, figures, observations, or information. It can be in the form of numbers, text, images, videos, or records.

Data is important because it helps us **understand situations, make better decisions, and find useful information**. Today, businesses use data to understand customers, hospitals use it to improve healthcare, and governments use it for planning and development.

Example: A shopping website collects data about what products people buy. This data helps the company understand customer preferences and recommend suitable products.

2) What is the difference between qualitative and quantitative data?

The main difference is that **qualitative data describes qualities or characteristics**, while **quantitative data is expressed in numbers**.

Qualitative Data	Quantitative Data
Describes qualities or opinions	Represents numerical values
Usually non-numerical	Numerical
Cannot usually be measured directly	Can be measured and calculated
Example: Colour, feedback, gender	Example: Age, height, marks, salary

Example:
If we ask students about their favourite subject, the answers like **Maths, Science, or Computer** are qualitative data.
If we record their **marks, age, or attendance percentage**, it is quantitative data.

3) What is data analysis, and how does it differ from data science?

Data analysis is the process of examining data to find useful information, patterns, and trends. It mainly helps in understanding **what has happened and why it happened**.

Data science is a broader field that includes data analysis along with **statistics, programming, machine learning, and predictive techniques**. It can be used to predict future outcomes and solve complex problems.

In simple words:

- **Data Analysis** → Understand and interpret existing data.

- **Data Science** → Analyze data and also use advanced methods to make predictions and build intelligent systems.

Example: A data analyst may study past sales to find which product sold the most, while a data scientist may use that data to predict future sales.

4) What are the key steps involved in a typical data analysis process?

The main steps in a data analysis process are:

1. **Define the Problem** – Understand what question needs to be answered.
2. **Collect Data** – Gather the required data from different sources.
3. **Clean the Data** – Remove errors, duplicate values, and handle missing data.
4. **Organize the Data** – Arrange the data in a suitable format for analysis.
5. **Analyze the Data** – Use statistical methods to find patterns and relationships.
6. **Visualize the Data** – Represent the results using graphs, charts, and tables.
7. **Interpret the Results** – Understand what the results mean.
8. **Make Decisions** – Use the findings to make better decisions or take suitable actions.

5) What tools and techniques are commonly used in data science?

Data science uses different **tools and techniques** to collect, process, analyze, and visualize data.

Common tools are:

- **Python** – Used for data analysis, machine learning, and programming.
- **R** – Mainly used for statistics and data analysis.
- **Excel** – Used for basic data analysis and visualization.
- **SQL** – Used to store, retrieve, and manage data from databases.
- **Power BI / Tableau** – Used to create charts, dashboards, and reports.

Common techniques are:

- Statistical analysis
- Data cleaning
- Data visualization
- Machine learning
- Data mining
- Predictive analysis
- Classification and clustering

These tools and techniques help data scientists **find useful patterns, make predictions, and support better decision-making**.